Where the dihedral mechanism stops: finitely ramified algebras, the threefold in $\mathcal{D}_{T_{18}}$, and the fields of finite groups on $\mathbb{P}^1$

Mumford algebras D over F

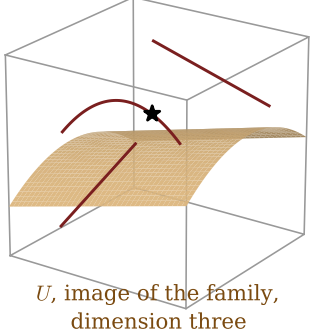
D_∞ D_{13} D_{29} D_{41} ...

ramified at two real places, and at two places above a split prime p

rank-twelve family, $\operatorname{disc}\Phi$ a square, only D_∞ is reached

(i)

$\mathcal{D}_{T_{18}}$, dimension four



U, image of the family, dimension three

Shimura curves of (F, D), not known to meet U

(ii)

finite groups acting on $\mathbb{P}^1$

C_m	$\mathbb{Q}(\zeta_m)$
D_m	$\mathbb{Q}(\zeta_m)^+$
A_4	$\mathbb{Q},\ \mathbb{Q}(\sqrt{-3})$
S_4	$\mathbb{Q}$
A_5	$\mathbb{Q},\ \mathbb{Q}(\sqrt{5})$

cubic only for D_7, D_9, D_{14}, D_{18}

(iii)